

SESSIONS 1:-

Q1.Name 5 IT job roles and write one line about what each role does ?

Ai/ml engineering:- Develops system that is use artificial intelligence and machine learning.
Web developer:- Create and maintains the website and web application.
Cloud engineer:-works with cloud platforms to deploy and manage applications.
Software developer:-Design and build application or software programs.
Network engineer:- Design and manages computer network and connectivity.

Q2.List 3 product companies and 3 service companies with one example of work each does?

Product companies:-

Google:- Search engine and Android OS.
Apple:-iphone and macOS.
Microsoft:- Window OS and like MS office.

Services companies:-

HCL tech:- Works for infrastructure and software services for compines.
TCS:-Developer solutions for client business.
Infosys:-It services to global cilents.

Q3.write the full forms and one-line of:-Github, Slack, jira, IDE, API?

Github:-Git+hub

Store ,manage and share code .

Slack:-Searchable log of all communication and knowledge
Communication and collaborations in companies.

Jira:-not full form after "gojira"

Track bugs, project, and tasks in software development.

IDE:- Integrated development environment.

Used by developers write , run and debug code.

API:- Application program interface

Allows different software applications to communication with each other.

Q4.Name any 5 programming languages and mention one popular use case for each ?

Python:- Data analysis and machine learning.
C :- System level programming.
Java:- Android app development.
C#:-games and developing windows.
Javascript:- web development.

Q5.What is the difference between a developer and a tester? Answer in 2 lines ?

Developer:-write and build software application or future.

Testers:-check the software for bugs and ensure it works correctly.

Sessions 2:-

Q1.What are the three main layers of software architecture? Write one line about each?

Presentation layer:-Manages user interface and interaction with the system.

Business Logic Layer :- Processes data and applies rules to perform core functions of the application.

Data Layer (Database):- Stores, retrieves, and manages data for the application

Q2. Define client and server in your own words with one real-life example?

Client:-A device or application that request services or data from another system.

Server:-A system that provides services or data in response to client request.

Ex:- open a website in your browser ,your browser is client and the web server send the webpage To you.

Q3. Write 2 differences each between desktop apps, web apps, and mobile apps?

Desktop apps:-

- Install directly on a computer .

- Work offline and system resource.

Web apps:-

- Run in web browser.

- Require internet connection to function properly.

Mobile apps:-

- Smart phone /tablets and install via app store.

- Use devices like camara,gps.

Q4. Name any 3 databases and mention whether each is SQL or NoSQL.

mySQL:- SQL database.

MongoDB:- NoSQL database.

PostgreSQL:- SQL database.

SESSIONS 3:-

Q1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

HTTP:- hyper text transfer protocols.

Used transfer webpage and data between browser and server.

HTTPS:- Hyper text transfer protocol secure.

Used securely data transfer.

FTP:- file transfer protocol.

Used upload and download files between computer over network.

SMTP:- Simple mail transfer protocol.

Used send email from one server to another .

Q2. What is the difference between IPv4 and IPv6? Answer in 2 lines?

IPv4:- uses 32-bit address and supports fewer devices.

IPv32:-uses 128-bit address and support a much larger number of devices.

Q3. Run nslookup google.com on your system and write down the IP address you get?

Run nslookup google.com get IP like 142.250.183.14 .

IP address change depending on your location and network result can be different .

Q4. What do these HTTP status codes mean: 200, 301, 404, 500, 403?

200(ok):- The request was successful.

301(Moved permanently):- resource been permanently moved to a new url.

404(Not found):- the request or resource was not found.

500(Internal server error):- The server encountered an error while processing the request.

403(Forbidden):- Access to the request resource is denied.

Q5. Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example?

Root(.):- The top of DNS it points to the correct TLD servers.

TLD(TOP LEVEL DOMAIN):- .COM category of domain.

Domain:- Main registered website name.

Subdomain:- A part of the domain used for specific services.

Flow : Root -> .com -> google.com -> www.google.com

SESSIONS :- 4

Q1.List all 6 phases of SDLC in order with one line about each?

Planning:- project goals, scope and feasibility.
Analysis:- Gather and analyze user requirements.
Design:- create architecture and design of the system.
Implementation:- develop the software by writing code.
Testing:- identify and fix bugs to ensure quality.
Deployment:- release the software to users.
Maintenance:- update the fix the system after release.

Q2. Write 3 differences between Agile and Waterfall models?

TOPIC	AGILE MODELS	WATERFALL MODELS
PROCESS	A follow iterative approach.	A follow a linear step-by-step approach
FLEXIBILITY	Flexible and allows change anytime.	Rigid with limited changes.
TESTING	Agile throughout development	Waterfall testing is done after development is completed.

Q3. Name the 3 main Scrum roles and write one line about the responsibility of each?

Product owner:- requirements and prioritizes the product backlog.

Scrum owner:- The team and ensures scrum practices are followed.

Development team:- builds and delivers the product increment.

Q4. What is a Sprint in Scrum? Mention its typical duration?

A sprint short, time boxed period in scrum where the team works to complete a set of tasks or deliver a product increment.

Typical duration :- 1 to 4 weeks

Q5. What do these Jira board columns mean: To Do, In Progress, Done, Backlog?

Jira backlog:- list of all tasks or issues that are planned but not yet started.

To do:- Tasks that are ready to be worked on.

In progress:- Tasks that are currently being worked on.

Done:- Tasks that have been completed.

SESSIONS 5:-

Q1. Write the Git command for each: initialize a repo, stage all files, commit with a message, push to remote, pull latest changes?

Initialize a repository :- git init.

Stage all files:- git add.

Commit with a message:- git commit -m "Your message".

Push to remote :- git push origin main.

Pull latest charge:- git pull origin main.

Q2. What is the difference between git fetch and git pull? Answer in 2 lines?

gitfetch :- The remote repository but does not merge them into your branch.

gitpull:-The change and automatically merge them into your current branch.

Q3. Create a new GitHub repository, push any small text file from your local system, and share the repo link?

Github-> "New repository"

Reponame-> click the create repository

Git init

Git add

Git status

Git commit -m "first commit"

Git branch -m main

Git remote add origin https://github.com/Rathodnimesh/software_engineering

Gitpush -u origin main

Q4. What is the difference between a Fork and a Pull Request? Answer in 2 lines?

Fork:- create a copy someone else repository in your own github account.

Pull request:- A request to merge your change from your repo into the original repository.

Q5. Create a new branch named feature-test, make one commit on it, and merge it back to the main branch. Write the commands you used?

Create and switch to new branch:- git check -b feature -test

Make a change and commit:- git add.

Git commit -m "Added feture test"

Switch back to backbranch:- git checkout main

Merge the branch into main:- git merge feature –test